DSA Day 22: BST - search & insert

Book: Data Structures Through C in Depth - Ch6 §6.12.1-6.12.4 · pp 202-207 | Code in Python

Overview

In a BST, left<root<right. Search and insert follow one root-to-leaf path.

Key concepts to master

- BST property
- O(h) search
- Insert as a leaf

Python implementation

```
def insert(root,v):
    if not root: return TNode(v)
    if v<root.data: root.left=insert(root.left,v)
    else: root.right=insert(root.right,v)
    return root
```

Complexity

Search/insert O(h): O(log n) balanced, O(n) skewed.

Common pitfalls

- Inserting duplicates without a rule
- Skewed tree degrading to O(n)

Practice

- Insert 50,30,70,20,40 and print inorder
- What makes a BST degrade?